

Revised Generation Interconnection System Impact Study Report

For Queue Project Q24-2447

Colfax 100 kV
October 24, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2447. The Interconnection Customer proposes a 69.5 MW battery energy storage facility interconnecting at the Colfax 100 kV substation in the Indiana area. The request was received on September 04, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 420 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,289,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Scenic Woods / Twin Branch Pv 100 kV Line
Contingency: Twin Branch / Freehaven-Lakepark Substation 100 kV L
Loading: 168.6% (Rating: 271 MVA)
DFAX: 21.1%
Pre-project loading: 163.2%
2. Facility: Segs 7 / Concord 100 kV Ckt 1
Contingency: Segs 7 / Concord 100 kV line
Loading: 137.3% (Rating: 512 MVA)
DFAX: 30.6%
Pre-project loading: 133.1%
3. Facility: Segs 7 / Concord 100 kV Ckt 2
Contingency: Concord / Segs 7 100 kV line
Loading: 137.3% (Rating: 512 MVA)
DFAX: 30.6%
Pre-project loading: 133.1%
4. Facility: Segs 7 / Freehaven-Lakepark 100 kV
Contingency: Segs 7 / Freehaven-Lakepark 100 kV #2
Loading: 134.6% (Rating: 543 MVA)
DFAX: 30.6%
Pre-project loading: 130.7%
5. Facility: Segs 7 / Freehaven-Lakepark 100 kV
Contingency: Freehaven-Lakepark / Segs 7 Substation 100 kV line
Loading: 134.6% (Rating: 543 MVA)
DFAX: 30.6%
Pre-project loading: 130.7%

6. Facility: Twin Branch / Freehaven-Lakepark 100 kV
Contingency: Whitestone / Stepstone 100 kV
Loading: 134.2% (Rating: 563 MVA)
DFAX: 31.1%
Pre-project loading: 130.4%
7. Facility: Carson County / Desert Shores 100 kV
Contingency: Table Mt / Gaston Rd 100 kV
Loading: 117.8% (Rating: 128 MVA)
DFAX: 6.8%
Pre-project loading: 114.1%
8. Facility: Kemper County / Dalreed 100 kV
Contingency: Gaston Rd / Table Mt 100 kV
Loading: 114.9% (Rating: 535 MVA)
DFAX: 70.4%
Pre-project loading: 105.7%
9. Facility: Desert Shores / Mimbres 100 kV
Contingency: Gaston Rd / Table Mt 100 kV Line
Loading: 114.3% (Rating: 123 MVA)
DFAX: 6.8%
Pre-project loading: 110.5%
10. Facility: Tabe Mt / Gaston Rd 100 kV
Contingency: Dalreed / Kemper County 100 kV
Loading: 112.0% (Rating: 611 MVA)
DFAX: 69.7%
Pre-project loading: 104.0%
11. Facility: Whitestone / Stepstone 100 kV
Contingency: Freehaven-Lakepark / Twin Branch 100 kV
Loading: 111.2% (Rating: 523 MVA)
DFAX: 16.5%
Pre-project loading: 109.0%
12. Facility: Whitestone / Molino 100 kV
Contingency: Freehaven-Lakepark / Twin Branch 100 kV
Loading: 111.2% (Rating: 505 MVA)
DFAX: 16.5%
Pre-project loading: 108.9%
13. Facility: Freehaven-Lakepark / Molino 100 kV
Contingency: Twin Branch / Freehaven-Lakepark 100 kV
Loading: 111.0% (Rating: 463 MVA)
DFAX: 16.5%
Pre-project loading: 108.5%
14. Facility: Northern Substation / Buchanan Mi 100 kV
Contingency: Table Mt / Gaston Rd 100 kV
Loading: 111.0% (Rating: 251 MVA)
DFAX: 22.6%
Pre-project loading: 104.8%
15. Facility: Bonnet Creek / Buchanan Mi 100 kV
Contingency: Table Mt / Gaston Rd Substation 100 kV L
Loading: 111.0% (Rating: 253 MVA)
DFAX: 22.6%
Pre-project loading: 104.7%
16. Facility: Ager Substation / Northern 100 kV
Contingency: Table Mt Substation / Gaston Rd 100 kV

- Loading: 108.6% (Rating: 282 MVA)
DFAX: 22.6%
Pre-project loading: 103.0%
17. Facility: Bennington / Grandview Ave 100 kV
Contingency: Gaston Rd / Table Mt 100 kV
Loading: 107.2% (Rating: 147 MVA)
DFAX: 14.7%
Pre-project loading: 100.3%
18. Facility: Bennington Substation / Grandview Ave 100 kV line
Contingency: Table Mt / Gaston Rd 100 kV
Loading: 107.2% (Rating: 147 MVA)
DFAX: 14.7%
Pre-project loading: 100.3%
19. Facility: Drewry / Colfax 100 kV
Contingency: Kemper County / Dalreed 100 kV
Loading: 106.4% (Rating: 202 MVA)
DFAX: 58.4%
Pre-project loading: 86.3%
20. Facility: Corridor / Larson Street 230 kV
Contingency: Trego / Beaver 500 kV
Loading: 102.4% (Rating: 1,567 MVA)
DFAX: 9.5%
Pre-project loading: 102.0%
21. Facility: Herald-Twin Cities Substation / Mason St 230 kV Line
Contingency: New Departure Substation / Mason St 230 kV Line
Loading: 102.2% (Rating: 1,155 MVA)
DFAX: 5.2%
Pre-project loading: 101.9%
22. Facility: N Mesquite / Denver Terminal 100 kV
Contingency: Twin Branch Pv / Parkrose 100 kV L
Loading: 101.9% (Rating: 264 MVA)
DFAX: 14.2%
Pre-project loading: 98.1%
23. Facility: Gaston Rd / Drewry 100 kV
Contingency: Dalreed / Kemper County 100 kV L
Loading: 101.8% (Rating: 408 MVA)
DFAX: 69.7%
Pre-project loading: 89.9%
24. Facility: Beaver Valley / Aes Beaver Valley Partners Beaver Valley 500 kV
Contingency: Cranberry / Beaver Vly Substation 500 kV Line
Loading: 100.8% (Rating: 3,206 MVA)
DFAX: 8.2%
Pre-project loading: 100.6%
25. Facility: Kemper County / Twin Branch 100 kV L
Contingency: Whitestone / Stepstone 100 kV
Loading: 100.2% (Rating: 599 MVA)
DFAX: 31.1%
Pre-project loading: 96.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Twin Branch Pv / Scenic Woods 100 kV	Rebuild line with higher-capacity conductor	\$1,109,000	4% (\$44,000)
Concord / Segs 7 Substation 100 kV Line 1	Rebuild line with higher-capacity conductor	\$1,933,000	4% (\$79,000)
Concord / Segs 7 100 kV Line 2	Rebuild line with higher-capacity conductor	\$2,603,000	4% (\$106,000)
Segs 7 / Freehaven-Lakepark 100 kV	Rebuild line with higher-capacity conductor	\$2,458,000	4% (\$100,000)
Segs 7 / Freehaven-Lakepark 100 kV	Rebuild line with higher-capacity conductor	\$3,706,000	4% (\$151,000)
Twin Branch / Freehaven-Lakepark 100 kV	Rebuild line with higher-capacity conductor	\$1,854,000	5% (\$85,000)
Carson County / Desert Shores 100 kV	Rebuild line with higher-capacity conductor	\$2,889,000	8% (\$239,000)
Kemper County / Dalreed 100 kV	Rebuild line with higher-capacity conductor	\$966,000	8% (\$73,000)
Desert Shores / Mimbres 100 kV	Rebuild line with higher-capacity conductor	\$9,044,000	8% (\$749,000)
Tabe Mt / Gaston Rd 100 kV	Rebuild line with higher-capacity conductor	\$1,456,000	8% (\$115,000)
Whitestone / Stepstone 100 kV	Rebuild line with higher-capacity conductor	\$1,907,000	3% (\$56,000)
Whitestone / Molino 100 kV	Rebuild line with higher-capacity conductor	\$3,154,000	3% (\$92,000)
Freehaven-Lakepark / Molino 100 kV	Rebuild line with higher-capacity conductor	\$967,000	3% (\$28,000)
Northern Substation / Buchanan Mi 100 kV	Rebuild line with higher-capacity conductor	\$6,953,000	6% (\$443,000)
Bonnet Crk / Buchanan Mi 100 kV	Rebuild line with higher-capacity conductor	\$2,959,000	6% (\$188,000)
Ager Substation / Northern 100 kV	Reconductor line	\$802,000	6% (\$51,000)
Bennington / Grandview Ave 100 kV L	Reconductor line	\$819,000	7% (\$57,000)
Bennington Substation / Grandview Ave 100 kV line	Reconductor line	\$683,000	7% (\$47,000)
Drewry / Colfax 100 kV	Reconductor line	\$973,000	21% (\$207,000)
Corridor / Larson Street 230 kV line	Reconductor line	\$5,878,000	0% (\$26,000)
Mason St / Herald-Twi Cities 230 kV L	Reconductor line	\$15,540,000	1% (\$103,000)
North Mesquite / Denver Terminal 100 kV line	Reconductor line	\$733,000	4% (\$33,000)
Drewry / Gaston Rd 100 kV line	Reconductor line	\$1,254,000	11% (\$139,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Aes Beaver Vly Partners Beaver Valley Substation / Beaver Valley Substation 500 kV	Reconductor line	\$8,603,000	0% (\$36,000)
Kemper County / Twin Branch 100 kV L	Reconductor line	\$915,000	5% (\$42,000)
	Total allocated to project		\$3,289,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.